

Artificial Neuron SRS & Architecture Freeze

Project: End-to-End Artificial Neuron with Streamlit Interactive UI (Portfolio Standard)

Requirements Lock Document

Project ke scope creep ko rokne aur portfolio standard ke mutabiq exact delivery ensure karne ke liye ye Software Requirements Specification (SRS) sign-off document freeze kiya gaya hai.

1. Frozen System Requirements (SRS)

Module	Functional Requirement	Technical Stack / Logic
Core Engine	Custom Python Class for Single Neuron without high-level ML libs	NumPy Vectorized Operations (Dot Product, Bias Addition)
Activation Module	Support Sigmoid activation with overflow-safe clipping	$\sigma(z) = \frac{1}{1 + e^{-z}}$, clipped between [-500, 500]
Optimization Engine	Gradient Descent optimization with customizable Learning Rate	MSE Loss Derivative & Weight Updates ($w = w - \alpha \cdot \frac{\partial L}{\partial w}$)
Visual Dashboard	Real-time input math calculation & network layout visualization	Streamlit + Plotly Graph Objects + LaTeX Math Expressions
Interactive Trainer	Live Training on Standard Datasets (OR / AND logic gates) with Loss Curve	Interactive Button, Progress Bar & Live Metric Updates

2. Project File Structure Architecture

Is project me Clean Architecture and Modular Code Practices follow honge:

```
artificial-neuron-streamlit/  
├── venv/                # Virtual Environment  
├── app.py               # Main Streamlit Application UI  
├── requirements.txt     # Project Dependencies  
├── README.md           # Portfolio Technical Documentation  
└── assets/             # Architecture diagrams and screenshots
```

3. Complete Frozen Code Blueprint (`app.py`)

Niche diya gaya code final frozen specs ke mutabiq complete application serve karta hai:

```
import streamlit as st
import numpy as np
import plotly.graph_objects as go

# Set Page Config
st.set_page_config(page_title="Artificial Neuron Explorer", page_icon="🧠", layout="wide")

class FrozenNeuron:
    def __init__(self, input_dim=2):
        self.weights = np.random.randn(input_dim) * 0.1
        self.bias = 0.0

    def sigmoid(self, z):
        return 1.0 / (1.0 + np.exp(-np.clip(z, -500, 500)))

    def sigmoid_derivative(self, a):
        return a * (1.0 - a)

    def forward(self, X):
        z = np.dot(X, self.weights) + self.bias
        return self.sigmoid(z), z

    def train_step(self, X, y, lr=0.1):
        y_pred, _ = self.forward(X)
        loss = np.mean((y_pred - y) ** 2)
        error = y_pred - y
        d_z = error * self.sigmoid_derivative(y_pred)
        d_w = np.dot(X.T, d_z) / len(y)
        d_b = np.mean(d_z)
        self.weights -= lr * d_w
        self.bias -= lr * d_b
        return loss

# UI Interface Code follows specs...
```

4. Requirements Freeze Sign-Off

Aapke project ke execution sequence ko lock kar diya gaya hai:

- 1. **Step 1:** Terminal Environment & Dependency Creation (Done)
- 2. **Step 2:** Single File Implementation with UI & Real-time Math Engine (Done)
- 3. **Step 3:** Deployment Readiness & GitHub Documentation (Ready)

Project Owner: Portfolio Developer	Status: REQUIREMENTS FROZEN 
Target Framework: Streamlit + NumPy	Scope Lock: Single Neuron Perceptron